

DECK 07 · WEEK 4 · AFTERNOON 1

FSA II — Cash Flow, Inventories & LT Assets

Three accounting choices that quietly drive earnings quality, tax, and comparability

CFA Level I · Vol 4 Learning Modules 4–7

Roadmap — FSA II: Cash Flow, Inventories & Long-Lived Assets

5 sections · ~2 hours · Built bottom-up from CFA readings

01

Cash flow statement · structure & preparation

3-bucket CFS · direct vs indirect · indirect→direct conversion · investing & financing build-up

02

Common-size CF + FCFF / FCFE

% inflows vs % revenue · FCFF from CFO and from NI · FCFE · 5 performance + 6 coverage ratios

03

Inventory methods & ratio impact

LCNRV · FIFO / LIFO / weighted-avg · inflation effects · LIFO reserve · LIFO liquidation · disclosures

04

PP&E · capitalize vs expense · depreciation

capitalize vs expense · SL / DDB / units · useful-life sensitivity · asset-age ratios from disclosures

05

Intangibles · impairment · R&D

purchased / internal / M&A · goodwill · impairment IFRS vs GAAP · derecognition · R&D IFRS vs GAAP

Learning objectives

By the end of this deck you should be able to:

1. Prepare operating cash flow under both the direct and indirect methods, reconcile them, and convert from one to the other using balance sheet and income statement data.
2. Compute FCFF and FCFE from both CFO and net income, and interpret cash-flow performance and coverage ratios.
3. Explain how FIFO, LIFO, and weighted-average cost flow assumptions — and the lower-of-cost-and-NRV rule — affect the income statement, balance sheet, and ratios under IFRS and US GAAP.
4. Decide whether an expenditure is capitalized or expensed, trace the consequences through income, assets, ratios and cash flow, and estimate asset age from disclosed cost and depreciation.
5. Account for intangibles by origin, test for impairment under IFRS vs US GAAP, and apply the distinct IFRS and US GAAP rules on research and development costs.

SECTION 01

Cash flow statement structure & preparation

The statement of cash flows reconciles opening and closing cash by sorting every movement into operating, investing, or financing

Ending cash = beginning cash + CFO + CFI + CFF + FX effects. Accrual profit is a judgement; the cash line is arithmetic.

$$\Delta \text{ Cash} = \text{CFO} + \text{CFI} + \text{CFF} \quad (+ \text{FX effect on cash})$$

OPERATING — CFO

Day-to-day business: receipts from customers, payments to suppliers, wages, taxes. Over the long run this is the only sustainable source of cash.

INVESTING — CFI

Buying and selling long-lived assets: PP&E, intangibles, investment securities. Capex lives here. Always presented gross.

FINANCING — CFF

Cash to and from capital providers: debt issued or repaid, equity issued or bought back, dividends paid. Also always gross.

WHERE IFRS GIVES YOU A CHOICE AND US GAAP DOES NOT

Interest paid — IFRS operating OR financing; US GAAP operating. Dividends paid — IFRS operating OR financing; US GAAP financing.
Interest and dividends received — IFRS operating OR investing; US GAAP operating.

Source: CFA Institute, Curriculum Vol 4, Learning Module 4, §1–2 and §8 Exhibit 34, pp. 124–127 and 146–147.

Direct and indirect methods produce the same CFO total — they only differ in how operating cash flow is presented

Both regimes encourage the direct method; almost every filer uses indirect. Analysts convert back — that is the next slide.

DIRECT METHOD — ACME 2018

Cash received from customers	23,543
Cash paid to suppliers	(11,900)
Cash paid to employees	(4,113)
Cash paid for other opex	(3,532)
Cash paid for interest	(258)
Cash paid for income tax	(1,134)

Net cash from operating activities	2,606
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Six gross lines. You can see the business.

INDIRECT METHOD — ACME 2018

Net income	2,210
Depreciation expense	1,052
Gain on sale of equipment	(205)
Increase in accounts receivable	(55)
Increase in inventory	(707)
Decrease in prepaid expenses	23
Increase in accounts payable	263
Other current-liability changes	25

Net cash from operating activities	2,606
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Source: CFA Institute, Curriculum Vol 4, Learning Module 4, §3–4, Exhibits 9–11 and 25, pp. 131–140.

Any indirect-method CFO can be re-expressed in direct format via a three-step process — a core analyst skill

The conversion is an approximation, but "sufficiently accurate for most analytical purposes". Every accrual line maps to one cash line.

Cash received from customers = Revenue – Δ Accounts receivable

CFA LM 4 EXHIBIT 26 — ACME CORP. 2018 CONVERSION

Revenue 23,598; COGS 11,456; salaries 4,123; depreciation 1,052; other opex 3,577; interest 246; tax 1,139; gain on sale 205. Δ AR +55, Δ Inv +707, Δ AP +263, Δ Prepaid –23, Δ WagesPay +10, Δ IntPay –12, Δ TaxPay +5, Δ Accrued +22.

1. Step 1 — Aggregate: total revenues 23,803 (23,598 + 205 gain); total expenses 21,593; net income 2,210.
2. Step 2 — Strip non-cash and non-operating: revenues 23,803 – 205 = 23,598; expenses 21,593 – 1,052 depreciation = 20,541.
3. Step 3 — Convert with Δ WC: customers 23,598 – 55 = 23,543; suppliers 11,456 + 707 – 263 = 11,900; employees 4,123 – 10 = 4,113; other opex 3,577 – 23 – 22 = 3,532.
4. Finish the same way: interest 246 + 12 = 258; income tax 1,139 – 5 = 1,134.

Direct CFO = 23,543 – 11,900 – 4,113 – 3,532 – 258 – 1,134 = USD 2,606 ✓

Source: CFA Institute, Curriculum Vol 4, Learning Module 4, §5 Exhibit 26, pp. 140–141.

A full cash flow statement walks from CFO through investing and financing to ending cash – and must tie to the balance sheet exactly

Investing and financing are always presented gross, regardless of which method is used for CFO. No netting, no shortcuts.

Line	Acme 2018	Section
Net cash from operating activities	2,606	CFO
Cash received from sale of equipment	762	CFI
Cash paid for purchase of equipment	(1,300)	CFI
Net cash used in investing activities	(538)	CFI
Cash paid to retire long-term debt	(500)	CFF
Cash paid to retire common stock	(600)	CFF
Cash paid for dividends	(1,120)	CFF
Net cash used in financing activities	(2,220)	CFF

THE TIE-OUT, AND THE THREE NUMBERS NOBODY GAVE YOU

Net decrease = $2,606 - 538 - 2,220 = (152)$. Opening cash $1,163 - 152 = 1,011$ — the closing balance sheet exactly. Debt retired 500 and stock retired 600 are simply year-on-year falls in those lines; dividends 1,120 come from the retained-earnings roll-forward, $2,876 + 2,210 - 3,966$.

Source: CFA Institute, Curriculum Vol 4, Learning Module 4, §4 Exhibit 25 and §7 Exhibit 32, pp. 139–140 and 145.

A net change in PP&E hides two gross flows — you need the income-statement gain to split purchases from disposals

Three roll-forwards, one answer: gross cost, accumulated depreciation, then book value plus the gain gives cash received.

$$\text{Cash from sale} = \text{Book value of asset sold} + \text{Gain on sale}$$

where: Book value = historical cost – accumulated depreciation, each recovered from its own balance-sheet roll-forward.

CFA LM 4 QUESTION SET – COPPER INC. EQUIPMENT DISPOSAL

Equipment 100 → 109. Accumulated depreciation 30 → 36. Capex USD 15m, all new equipment. Income statement: depreciation 8m, gain on sale 12m. How much cash came in?

1. Historical cost of equipment sold: $100 + 15 - 109 = \text{USD } 6\text{m}$
2. Accumulated depreciation on equipment sold: $30 + 8 - 36 = \text{USD } 2\text{m}$
3. Book value of equipment sold: $6 - 2 = \text{USD } 4\text{m}$
4. Cash received: book value + gain = $4 + 12 = \text{USD } 16\text{m}$

CFI shows +16 inflow and -15 outflow, not a single net line of +1.

SECTION 02

Common-size & free cash flow

Common-size cash flow analysis scales each line either to total inflows and outflows or to net revenue – the two answer different questions

Acme 2018: total inflows USD 24,305, total outflows USD 24,457, net change (152). Revenue base for the right-hand column is net revenue of USD 23,598.

Acme 2018 – direct-format cash flow line	Amount (USD)	% inflows / outflows	% net revenue
Cash received from customers	23,543	96.86%	99.77%
Cash received from sale of equipment	762	3.14%	3.23%
Cash paid to suppliers	(11,900)	48.66%	50.43%
Cash paid to employees	(4,113)	16.82%	17.43%
Cash paid for other operating expenses	(3,532)	14.44%	14.97%
Cash paid for interest and income tax	(1,392)	5.69%	5.90%
Purchase of equipment	(1,300)	5.32%	5.51%
Retirement of long-term debt and stock	(1,100)	4.50%	4.66%
Dividends paid	(1,120)	4.58%	4.75%

WHY BOTH, NOT ONE

The inflow view asks where the cash comes from. The revenue view asks whether each item scales with sales.

Source: CFA Institute, Curriculum Vol 4, Learning Module 5, §3 Exhibits 4–5, pp. 161–163.

FCFF and FCFE separate the cash available to all capital providers from the cash available to equity holders alone

Discount FCFF at WACC to value the firm; discount FCFE at the cost of equity to value the equity. Mismatching the pair is the classic valuation error.

$$\text{FCFF} = \text{NI} + \text{NCC} + \text{Int}(1-t) - \text{FCInv} - \text{WCInv}$$

$$\text{FCFF} = \text{CFO} + \text{Int}(1-t) - \text{FCInv}$$

$$\text{FCFE} = \text{CFO} - \text{FCInv} + \text{Net borrowing}$$

where: NCC = non-cash charges (D&A). FCInv = net capex. WCInv = rise in working capital. Net borrowing = issued – repaid.

FCFF – FIRM LEVEL

Cash for ALL capital providers after operating costs, taxes and reinvestment in fixed and working capital.

Interest is added back AFTER tax because debt-holders were already paid out of this cash — and because the interest deduction saved real tax.

FCFE – EQUITY LEVEL

Cash for common shareholders after interest, principal and reinvestment.

Do NOT add back interest. DO add net borrowing — and remember it is NEGATIVE when debt is repaid, which is where the sign errors live.

Acme's FCFF and FCFE differ by USD 670 — after-tax interest going one way and debt repayment the other

Tax rate is derived, not given: $1,139 \text{ tax expense} \div 3,349 \text{ pretax income} = 34\%$. Same CFO of 2,606 feeds both measures.

CFA LM 5 EXHIBITS 7 & 8 — ACME CORP. FCFF AND FCFE

CFO 2,606. Capex 1,300. Proceeds from equipment sale 762. Interest paid 258. Long-term debt retired 500, none issued. Tax rate $t = 34\%$.

1. FCInv (net) = capex 1,300 – disposal proceeds 762 = USD 538
2. FCFF = CFO + Int(1-t) – FCInv = $2,606 + 258 \times 0.66 - 538 = 2,606 + 170 - 538 = \text{USD } 2,238$
3. Net borrowing = 0 issued – 500 repaid = USD (500)
4. FCFE = CFO – FCInv + Net borrowing = $2,606 - 538 - 500 = \text{USD } 1,568$

FCFF 2,238 · FCFE 1,568 · gap 670 = 170 interest add-back + 500 repayment

CROSS-CHECK FROM NET INCOME — AND WHAT IT MEANS

NI 2,210 + NCC 1,052 – gain 205 – WCInv 451 = 2,606 = CFO. Then – FCInv 538 – net borrowing 500 = 1,568. Same answer, different road. WCInv 451 = operating current assets up 739 (AR 55, inventory 707, prepaid –23) less operating current liabilities up 288. FCFE 1,568 against a dividend of 1,120 is 1.40× cover: the payout is funded, but only just, and only after a year of net deleveraging.

Cash flow ratios split into five performance ratios that measure profitability and six coverage ratios that measure solvency

Every one of them has CFO in the numerator. That is the point: cash-based ratios sidestep accrual judgement entirely.

Ratio	Formula	What it measures	Type
Cash flow to revenue	$\text{CFO} / \text{Net revenue}$	Cash per unit of revenue	Perf.
Cash return on assets	$\text{CFO} / \text{Avg. total assets}$	Cash per unit of assets	Perf.
Cash return on equity	$\text{CFO} / \text{Avg. shareholders' equity}$	Cash per unit of equity	Perf.
Cash to income	$\text{CFO} / \text{Operating income}$	Earnings quality	Perf.
Cash flow per share	$(\text{CFO} - \text{Pref. div.}) / \text{Common shares}$	Per-share operating cash	Perf.
Debt coverage	$\text{CFO} / \text{Total debt}$	Financial risk / leverage	Cover.
Interest coverage	$(\text{CFO} + \text{Int} + \text{Tax paid}) / \text{Int paid}$	Can it service its debt?	Cover.
Reinvestment	$\text{CFO} / \text{Cash paid for LT assets}$	Can CFO fund capex?	Cover.
Debt payment	$\text{CFO} / \text{Cash paid for debt repayment}$	Can CFO repay principal?	Cover.
Dividend payment	$\text{CFO} / \text{Dividends paid}$	Can CFO fund dividends?	Cover.
Investing and financing	$\text{CFO} / \text{Investing} + \text{financing outflows}$	Can CFO fund all outflows?	Cover.

Source: CFA Institute, Curriculum Vol 4, Learning Module 5, §5 Exhibit 9, pp. 168–169.

Read cash-flow ratios as a trend, not a level — and check what changed in the denominator before you call it earnings quality

Both are exceptional cash generators. The interesting part is that the two ratios tell different stories about the same three years.

Metric	Company	2015	2016	2017	Change 2015 → 2017
CFO / net revenue	Microsoft	31.7%	39.1%	43.9%	+12.2 pp
CFO / net revenue	Apple Inc.	34.8%	30.5%	27.7%	-7.1 pp
CFO / avg. total assets	Microsoft	17.1%	18.1%	18.2%	+1.1 pp
CFO / avg. total assets	Apple Inc.	31.1%	21.5%	18.2%	-12.9 pp

WHAT THE NUMBERS ACTUALLY SAY

1. CFO / revenue — Microsoft converts 43.9 cents of every sales dollar into operating cash, versus 27.7 at Apple. Microsoft is rising, Apple falling; on this measure the ranking reversed between 2015 and 2016.
2. CFO / assets — Apple's collapse from 31% to 18% is NOT an operating deterioration. The denominator ballooned with a marketable-securities portfolio built out of past cash flow. Same numerator quality, different balance sheet.
3. The discipline: before calling a falling ratio a warning, ask which half of it moved and why.

Source: CFA Institute, Curriculum Vol 4, Learning Module 5, §5 Example 3, Exhibits 10–11, p. 169.

SECTION 03

Inventory methods & ratio impact

IFRS carries inventory at the lower of cost and NRV; US GAAP does the same except for LIFO and retail filers — and never permits a reversal

"Market" is a US GAAP concept and applies only to LIFO and retail filers: replacement cost, capped at NRV and floored at NRV less a normal profit margin.

$$\text{Carrying amount} = \text{MIN (cost , net realisable value)}$$

where: NRV = selling price in the ordinary course of business – costs to complete – costs to sell. Assessed item by item.

Question	IFRS (IAS 2)	US GAAP
Measurement base	Lower of cost and NRV	Lower of cost and NRV
...except LIFO / retail filers	n/a – LIFO is prohibited	Lower of cost or market
Where the write-down lands	Cost of sales (or separate line)	Cost of sales
Reversal if value recovers	Required, capped at original cost	Prohibited, permanently
Effect on activity ratios	Turnover looks better	Turnover looks better

THE RATIO ASYMMETRY WORTH REMEMBERING

A write-down cuts both profit and inventory, so profitability, liquidity and solvency ratios all worsen — but activity ratios IMPROVE, because the denominator shrank. Never read that improvement as better inventory management.

Source: CFA Institute, Curriculum Vol 4, Learning Module 6, §2, pp. 174–176.

FIFO, LIFO and weighted average split the same cost pool differently between COGS and ending inventory — same purchases, different profit and different tax

The physical flow of goods is irrelevant to the choice. LIFO is not permitted under IFRS; FIFO and weighted average are permitted under both regimes.

F I F O

First-in, first-out

Oldest costs → COGS

Newest costs → ending inventory

Balance sheet is close to replacement cost.

Income statement matches old costs to current sales.

RISING PRICES: low COGS, high profit, high tax

FALLING PRICES: everything reverses

L I F O (U S G A A P o n l y)

Last-in, first-out

Newest costs → COGS

Oldest costs → ending inventory

Income statement is close to replacement cost.

Balance sheet carries stale, low costs.

RISING PRICES: high COGS, low profit, LOW TAX

FALLING PRICES: everything reverses

W E I G H T E D A V E R A G E

One blended cost per unit

Avg cost = cost of goods available ÷ units available

Every unit, in COGS and on the balance sheet, carries the same cost.

Always sits BETWEEN FIFO and LIFO on every metric — which makes it the useful sanity check.

Source: CFA Institute, Curriculum Vol 4, Learning Module 6, §3, pp. 181–182.

In a rising-price environment LIFO raises COGS and freezes ending inventory – cutting reported income and tax, and quietly distorting the ratios

Companies L and F are identical except for the cost-flow assumption. 4% annual price inflation, 10% annual unit growth, a 2,000-unit base layer bought at USD 8.

Year	Ending inv. (L)	Ending inv. (F)	COGS (L)	COGS (F)	Gross profit (L)	Gross profit (F)
1	16,000	16,000	160,000	160,000	140,000	140,000
2	16,000	16,640	183,040	182,400	160,160	160,800
3	16,000	17,306	209,398	208,732	183,223	183,889
4	16,000	17,998	239,551	238,859	209,607	210,299
5	16,000	18,718	274,046	273,326	239,791	240,511
Total	–	+2,718 vs L	1,066,035	1,063,317	932,781	935,499

THE SAME 2,718 SHOWS UP FOUR TIMES

1. Inventory — F ends year 5 at 18,718 against L's 16,000. The 2,718 gap is inflation, not more stock.
2. COGS — L charged 2,718 more over five years, so gross profit is 2,718 lower. Same pool, different split.
3. Turnover (COGS ÷ ending inventory) — L climbs 10.0× to 17.1×, F only 10.0× to 14.6×. LIFO merely looks better run.
4. Tax — at 25%, L's lower gross profit defers roughly USD 680 of cash tax. That is LIFO's entire case.

Source: CFA Institute, Curriculum Vol 4, Learning Module 6, §3 Example 3, pp. 182–183.

The LIFO reserve is a disclosed footnote that bridges a US LIFO filer back to a comparable FIFO view

LIFO filers must disclose the FIFO equivalent. ZP, a Japan-domiciled US GAAP filer, reports a reserve of JPY 10,120m in 2017 rising to JPY 19,660m in 2018.

$$\begin{aligned}\text{Inventory (FIFO)} &= \text{Inventory (LIFO)} + \text{LIFO reserve} \\ \text{COGS (FIFO)} &= \text{COGS (LIFO)} - \Delta \text{LIFO reserve}\end{aligned}$$

ZP Corporation (JPY millions)	As reported (LIFO)	LIFO reserve	Restated to FIFO
Inventory, 31 December 2017	608,572	10,120	618,692
Inventory, 31 December 2018	486,465	19,660	506,125
Change in LIFO reserve during 2018	—	+9,540	—
Cost of products sold, 2018	5,822,805	– 9,540	5,813,265
Inventory turnover, 2018	10.64×		10.34×

BOTTOM LINE

Restating to FIFO lifts inventory and trims COGS, so turnover FALLS from 10.64× to 10.34× and the current ratio rises. A LIFO filer is not better at managing inventory — it is carrying old costs. Under IFRS the choice does not exist.

Source: CFA Institute, Curriculum Vol 4, Learning Module 6, Practice Problems 28–33 (ZP Corporation), pp. 200–204.

A LIFO liquidation — selling more than you replace — releases old low-cost layers into COGS and inflates gross profit unsustainably

A falling inventory balance at a LIFO filer is a red flag. US GAAP requires disclosure of any material income arising from a LIFO liquidation.

STYLISTED — EXTENDING COMPANY L'S BASE LAYER

Company L still holds its 2,000-unit base layer at the original USD 8. Current purchase cost has reached USD 14. This year it sells 2,500 units but buys only 2,000.

1. COGS if every unit came from this year's purchases: $2,500 \times 14 = \text{USD } 35,000$.
2. Actual LIFO COGS: $2,000 \times 14$ (this year's layer) + 500×8 (base layer) = $28,000 + 4,000 = \text{USD } 32,000$.
3. Liquidation gain: $35,000 - 32,000 = \text{USD } 3,000$, i.e. 500 units $\times$ the USD 6 spread between current and base cost.
4. Ending inventory falls by $500 \times 8 = \text{USD } 4,000$, and the LIFO reserve shrinks by the same 3,000 of released spread.

Analyst adjustment: strip the 3,000 out before judging sustainable gross margin.

WHY IT HAPPENS — AND WHY IT IS EXPENSIVE

A liquidation can be involuntary (a strike, a supply shock) or engineered — run inventory down and book a one-off margin uplift in a weak quarter. Either way the earnings boost is real cash tax paid: the liquidation unwinds precisely the deferral the LIFO reserve was holding. Management is trading tomorrow's tax shield for today's EPS.

Source: CFA Institute, Curriculum Vol 4, Learning Module 6, §3 Example 3 note and §4, pp. 182 and 184.

Inventory disclosures let you strip the obsolescence allowance out and see what turnover and margin really look like

Volvo carries inventory at FIFO, net of an obsolescence allowance of SEK 3,489m at end-2017. The allowance FELL 194 during the year, which flattered cost of sales.

Volvo Group 2017	As reported	Without allowance	Why it moves
Inventory, net (SEK m)	52,701	56,190	Add back the 3,489 allowance
Cost of sales (SEK m)	254,581	254,775	Allowance fell 194 in 2017
Inventory turnover	5.04×	4.71×	Write-downs shrink the denominator
Days of inventory on hand	72.4	77.5	$365 \div \text{inventory turnover}$
Gross profit margin	23.95%	23.89%	Lower COGS as reported
Net profit margin	6.36%	6.31%	Tax at 25% damps the gap

ANALYST CHECKLIST

1. Read the policy note first — FIFO, weighted average or LIFO changes every ratio below it.
2. If LIFO, pull the reserve and restate to FIFO before any cross-border comparison.
3. Add the obsolescence allowance back to see gross inventory; a rising allowance is an obsolescence signal.
4. Compare inventory growth with revenue growth. Inventory outrunning sales is the classic warning.

Source: CFA Institute, Curriculum Vol 4, Learning Module 6, §2 Example 2 (Volvo Group), pp. 177–180.

One write-down, one recovery, two regimes — and a permanent EUR 0.5m gap between an IFRS filer and its US GAAP peer

Hatsumei manufactures computers and reports under IFRS. The comparison assumes an otherwise identical US GAAP filer using LIFO, for which "market" applies.

CFA LM 6 EXAMPLE 1 — HATSUMEI ENTERPRISES

2017: ending inventory cost EUR 5.2m; NRV EUR 4.9m; current replacement cost EUR 4.7m. 2018: NRV recovers to EUR 0.5m above the carrying amount.

1. IFRS 2017 — carry at $\text{MIN}(\text{cost } 5.2, \text{NRV } 4.9) = 4.9$. The 0.3m decline is expensed.
2. IFRS 2018 — reverse, capped at the original 0.3m: inventory back to 5.2, cost of sales cut by 0.3m.
3. US GAAP 2017 (LIFO filer) — market = replacement cost 4.7, below NRV 4.9. Write down to 4.7; 0.5m into COGS.
4. US GAAP 2018 — no reversal, ever. Inventory sits at 4.7 while the IFRS peer carries 5.2.

Two-year P&L hit: IFRS zero (−0.3 then +0.3). US GAAP −0.5m, permanently.

THE THIRD ANSWER MOST STUDENTS MISS

If the inventory were an agricultural product rather than computers, IAS 2 would not apply at all: it would be carried at NRV and the 2018 recovery booked as a EUR 0.5m GAIN — not a reversal capped at original cost.

Source: CFA Institute, Curriculum Vol 4, Learning Module 6, §2 Example 1 (Hatsumei Enterprises), pp. 175–176.

SECTION 04

PP&E • capitalize vs expense • depreciation

Capitalizing spreads a cost over future periods; expensing takes it today – and the choice moves income, assets, ratios and even which cash flow line it lands in

Capitalize when the expenditure delivers benefit beyond the current period and can be measured reliably. Otherwise expense as incurred.

In the year of the spend	Capitalize	Expense
Net income / net margin	Higher – only depreciation hits	Lower – the full cost hits
Total assets	Higher – an asset is recorded	Lower – nothing recorded
Shareholders' equity	Higher	Lower
ROA and ROE	Higher – income effect dominates	Lower
CFO	Higher – outflow sits in CFI	Lower – outflow inside CFO
Total asset turnover	Lower – bigger denominator	Higher
Debt to equity	Lower	Higher
Later-year profit GROWTH	Lower – depreciation drags	Higher – no charge left

CAPITALISED INTEREST – THE ONE THAT CATCHES PEOPLE

Capitalised interest lands in PP&E, not in interest expense, so fixed-asset turnover falls. Interest coverage only looks better if you compute it on interest EXPENSE — build it on total interest PAID, capitalised portion included, and the flattery disappears.

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §2 p. 213 and Practice Problems 4 and 8–9, pp. 234–237.

Straight-line is the default; double-declining balance front-loads the charge; units-of-production ties it to actual use

All three write off the SAME total (cost – salvage) over the same life. They differ only in timing — and therefore only in the shape of the reported margin.

STRAIGHT - LINE

$\text{Dep} = (\text{Cost} - \text{Salvage}) \div \text{Useful life}$

Flat annual charge.

Net profit margin: the FLATTEST trend of the three.

Use where benefits are even across the life — buildings, fittings, most office assets.

DOUBLE - DECLINING BALANCE

$\text{Dep} = (2 \div \text{Useful life}) \times \text{opening NBV}$

Accelerated — biggest charge in year 1.

Salvage does NOT enter the formula; you simply stop once NBV reaches it.

Net profit margin: IMPROVES year on year as the charge falls.

UNITS OF PRODUCTION

$\text{Dep} = (\text{Cost} - \text{Salvage}) \times (\text{Units used} \div \text{Total units})$

Activity-based — matches wear to output.

Net profit margin: VARIABLE, moving with volume.

Use where consumption is measurable — mine tonnes, aircraft cycles, press impressions.

Source: CFA Institute, Curriculum Vol 4, Learning Module 7 — assumed foundation, tested in Practice Problems 5 and 10, pp. 235 and 242.

A USD 50,000 asset, five-year life, USD 5,000 salvage — identical total depreciation, completely different reported curve

DDB rate = $2 \div 5 = 40\%$, applied to opening net book value. Salvage is ignored in the formula — you simply stop depreciating once NBV reaches it.

Year	SL expense	SL net book value	DDB expense	DDB net book value
1	9,000	41,000	20,000	30,000
2	9,000	32,000	12,000	18,000
3	9,000	23,000	7,200	10,800
4	9,000	14,000	4,320	6,480
5	9,000	5,000	1,480	5,000
Total	45,000	Ends at salvage	45,000	Ends at salvage

THE TAX POINT — AND ITS CAVEAT

Year 1 at a 25% rate: SL cuts pretax income by 9,000 (tax 2,250); DDB cuts it by 20,000 (tax 5,000) — 2,750 of cash tax deferred. Over five years both charge 45,000, so total tax is identical. DDB is timing, not value.

Caveat: this only bites where tax depreciation follows the book method. Where it does not — most of Europe — the choice moves deferred tax, not cash.

Source: CFA Institute, Curriculum Vol 4, Learning Module 7 — illustration of Practice Problems 5 and 10, pp. 235 and 242.

Useful life, salvage value and method are three estimates, not three facts — and small revisions produce large earnings effects

Same USD 50,000 asset throughout. Stretching the useful life from five years to eight cuts the annual charge by 38% — with no change whatsoever in the asset.

Assumption change	Annual SL depreciation	Δ vs baseline	Year-5 NBV
Baseline: life 5y, salvage 5,000	9,000	—	5,000
Life extended to 8y (same salvage)	5,625	-3,375 (-38%)	21,875
Salvage raised to 10,000 (life still 5y)	8,000	-1,000 (-11%)	10,000
Switch to DDB (life 5y, salvage 5,000)	20,000 in yr 1	+11,000 yr 1	5,000
Both: life 8y AND salvage 10,000	5,000	-4,000 (-44%)	25,000

HOW TO VET THE ESTIMATES

1. IFRS requires disclosure of useful lives or depreciation rates by class of asset. Benchmark them against three peers, not against the company's own history.
2. Watch for an upward revision to useful life announced alongside a weak quarter. Raising life AND salvage together is a double-barrelled earnings boost.
3. The next slide gives you the counter-test: back out the implied life from the disclosed cost and depreciation expense, and see whether it matches what the company says it uses.

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §4 disclosure requirements, p. 223; sensitivity computed on the slide-27 asset.

Gross cost, accumulated depreciation and depreciation expense are enough to estimate how old an asset base is – and when the capex bill lands

Three telecoms, one year, three depreciation policies. Orange and BCE write PP&E off over 21–23 years; Verizon over 17. Roughly two-thirds of every base is already consumed.

$$\begin{aligned}\text{Age} &= \text{Accum. depreciation} \div \text{Dep. expense} \\ \text{Remaining life} &= \text{Net PP\&E} \div \text{Dep. expense}\end{aligned}$$

where: Total useful life = age + remaining life = gross PP&E ÷ dep. expense. Straight-line assumed — an indicator, not a measure.

2017, PP&E under the cost model	Orange (EUR m)	BCE (CAD m)	Verizon (USD m)
Gross PP&E	97,092	69,230	246,498
Accumulated depreciation	70,427	45,197	157,930
Annual depreciation expense	4,708	3,037	14,741
⇒ Estimated total useful life	20.6 yrs	22.8 yrs	16.7 yrs
⇒ Estimated age	15.0 yrs	14.9 yrs	10.7 yrs
⇒ Estimated remaining life	5.7 yrs	7.9 yrs	6.0 yrs
Capex ÷ dep. · FA turnover	1.2x · 1.6x	1.4x · 1.0x	1.2x · 1.5x

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §5 Example 6 (Orange / BCE / Verizon), pp. 229–232.

SECTION 05

Intangibles • impairment • R&D

How an intangible arrived decides how it is accounted for — purchased, internally developed, or acquired in a business combination

Goodwill exists only in a business combination. It is the residual after every identifiable asset and liability has been fair-valued — never something you can buy on its own.

PURCHASED SEPARATELY

A patent, a licence, a domain name bought from a third party.

- Recorded at fair value = purchase price
- Buy a group? Allocate the price across them by relative fair value
- Finite life: amortise, test on trigger
- Indefinite life: no amortisation, annual test

INTERNALLY DEVELOPED

Generally EXPENSED as incurred.

- IFRS: research expensed; development capitalised once the criteria are met
- US GAAP: R&D expensed, except certain software

The result is a structural asymmetry: build it and it never appears; buy it and it does.

ACQUIRED IN M & A

Acquisition method: allocate the purchase price to each asset and liability at fair value.

- Identifiable intangibles get their own lines — brands, patents, customer lists
- The residual is GOODWILL
- Not amortised; tested at least annually

¹ Cash flow trap: internally developing an intangible is an OPERATING outflow; buying the same intangible is an INVESTING outflow.

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §2, pp. 212–215.

Impairment writes an asset down when carrying amount exceeds recoverable amount — and IFRS and US GAAP can give opposite answers on the same machine

Two almost identical machines. On Sussex, US GAAP records nothing while IFRS records GBP 2,000. On Essex, US GAAP records double the IFRS loss.

$$\text{IFRS loss} = \text{CA} - \text{MAX} (\text{FV} - \text{costs to sell} , \text{value in use})$$

where: US GAAP, step 1: is CA above undiscounted cash flows? Only then, step 2: loss = CA – fair value. IFRS does it in one step.

Same machine, two fact patterns (GBP)	Sussex — Example 2	Essex — Example 3
Carrying amount	18,000	18,000
Undiscounted expected future cash flows	19,000	16,000
Value in use (PV of cash flows)	16,000	14,000
Fair value less costs to sell	17,000 – 2,000 = 15,000	10,000 – 2,000 = 8,000
IFRS: CA – MAX(FV–sell, VIU)	Loss 2,000 → carry 16,000	Loss 4,000 → carry 14,000
US GAAP step 1: CA vs undisc. CF	18,000 < 19,000 → OK	18,000 > 16,000 → fails
US GAAP step 2: loss = CA – FV	No loss at all	Loss 8,000 → carry 10,000

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §3 Examples 2 and 3 (Sussex, Essex), pp. 217–219.

Derecognition removes an asset from the balance sheet — gain or loss is simply proceeds minus carrying amount

Under the indirect method the GAIN is removed from CFO and the FULL proceeds appear in CFI. Leave the gain in and you have counted the same cash twice.

$$\text{Gain / (loss) on sale} = \text{Sale proceeds} - \text{Carrying amount}$$

where: Carrying amount = historical cost – accumulated depreciation – any prior impairment, or the revalued amount under the IFRS revaluation model.

CFA LM 7 EXAMPLE 4 — MOUSSILAUKE DINERS SELLS 450 PIZZA OVENS

Historical cost USD 5.1m. Accumulated depreciation USD 3.2m. Sale proceeds USD 3.1m. The chain is repositioning its menu towards healthier items.

1. Carrying amount: $5.1 - 3.2 = \text{USD } 1.9\text{m}$
2. Gain on sale: $3.1 - 1.9 = \text{USD } 1.2\text{m}$
3. Income statement: 1.2m gain, usually inside other gains and losses.
4. Cash flow statement (indirect): subtract 1.2m from CFO, show the full 3.1m as a CFI inflow.

Gain 1.2m in P&L • +3.1m in investing • zero in operating

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §3 Example 4 (Moussilauke Diners), pp. 220–221.

IFRS capitalises development costs once six criteria are met; US GAAP expenses all R&D except certain software

IFRS criteria: technical feasibility, intent to complete, ability to use or sell, probable future benefits, adequate resources, reliable measurement of the cost.

CFA LM 7 EXAMPLE 1 — REH AG SOFTWARE FOR INTERNAL USE

REH AG reports under IFRS and spends EUR 1,000 a month through the year ended 31 December 2019 developing software for internal use. Annual total EUR 12,000.

1. Criteria met on 1 February: January's EUR 1,000 is expensed; EUR 11,000 is capitalised as an intangible asset.
2. Criteria met on 1 December: EUR 11,000 of January-to-November spend is expensed; only EUR 1,000 is capitalised.
3. Under US GAAP, with completion already probable in 2018: the full EUR 12,000 is capitalised as internal-use software.
4. Costs already expensed can never be reclassified to an asset later. The date the criteria are met is decisive.

IFRS is date-sensitive; US GAAP software is probability-sensitive.

WHY THIS MAKES PEERS INCOMPARABLE

A capitaliser reports higher profit, higher assets and higher CFO than an identical expenser — three ratios distorted at once, and the cash outflow moves from operating to investing. The standard fix is to expense ALL development spend across the peer group before ranking anything.

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §2 Example 1 (REH AG), pp. 214–215.

Goodwill is the residual in an acquisition – not amortised, tested at least annually, and on a large deal it can be most of the price

AB InBev paid USD 103.1bn for SABMiller in 2016 and recognised USD 74.1bn of goodwill. Brands with indefinite life account for USD 19.9bn of the USD 20.0bn intangibles line.

AB InBev provisional purchase-price allocation	USD million	% of price
Property, plant & equipment	9,060	8.8%
Intangible assets (brands, licences, customer lists)	20,040	19.4%
Associates, inventories, receivables, cash, other	9,117	8.8%
Assets held for sale (MillerCoors, Miller, CR Snow...)	24,805	24.1%
Less: total liabilities and non-controlling interests	(33,969)	(32.9)%
Net identifiable assets acquired	29,053	28.2%
Goodwill on acquisition	74,083	71.8%
Total purchase consideration	103,136	100.0%

WHERE THE IMPAIRMENT SHOWS UP – AND WHAT DRIVES IT

Orange SA booked goodwill impairments of EUR 772m in 2016 and EUR 20m in 2017, and disclosed the assumptions behind them: post-tax discount rates from 5.5% in France to 8.8% in Romania, perpetuity growth 0.5% to 2.3%. Read those inputs before you trust any goodwill carrying amount.

Source: CFA Institute, Curriculum Vol 4, Learning Module 7, §2 Exhibit 1 (AB InBev / SABMiller) p. 216, and §4 Example 5 (Orange SA) pp. 225–226.

Deck 7 — FSA II: Cash Flow, Inventories & Long-Lived Assets

- CFO is identical under direct and indirect. Indirect→direct is three steps: aggregate, strip non-cash, adjust ΔWC . Acme 2018 = USD 2,606 both ways.

- $FCFF = CFO + Int(1-t) - FCInv$. $FCFE = CFO - FCInv + \text{net borrowing}$. Acme: 2,238 and 1,568 — a 670 gap, all interest and repayment.

- Rising prices: LIFO gives higher COGS, lower income, lower tax, frozen inventory. $Inv(FIFO) = Inv(LIFO) + \text{reserve}$; $COGS(FIFO) = COGS(LIFO) - \Delta \text{ reserve}$.

- Capitalising raises year-1 income, assets, ROA and CFO, and lowers asset turnover. Over the full life cumulative income is identical.

- Sign rule: operating ASSET up = cash down; operating LIABILITY up = cash up. Investing and financing are always gross.

- Cash flow ratios: five performance (revenue, assets, equity, income, per share) and six coverage (debt, interest, reinvestment, debt payment, dividend, all outflows).

- LCNRV: write-downs hit COGS and make activity ratios look BETTER. IFRS reverses up to original cost; US GAAP never does. Hatsumei: 5.2 vs 4.7.

- Impairment: $IFRS = CA - \text{MAX}(FV - \text{sell}, VIU)$, one step. US GAAP screens on undiscounted cash flows, then measures to fair value. Essex: 4,000 vs 8,000.